

4a	work done per unit positive charge in bringing a small test charge from infinity to that point
bi	Gradient of the V-r graph
bii	Horizontal line showing $E = 0$ within the conductor (at right and left sides of graph) *has to be drawn even though it is at zero Shape of graph with a turning point (min in the +ve segment) Graph is asymmetrical with turning point nearer to left side, with the left side of the curve being higher than the right side.
biii	Since $V_P + V_Q = 0$ $\frac{3.2 \times 10^{-9}}{4\pi\epsilon_0 x} + \frac{-1.6 \times 10^{-9}}{4\pi\epsilon_0 (0.40 - x)} = 0$ $x = 0.27 \text{ m}$
biv	V at all points doubled Position when $V = 0$ remains unchanged
ci1	$W_{AB} \text{ (by electric force)} = 5.0 \times 10^{-7} \times 2.0 \times 10^{-2} = 1.0 \times 10^{-8} \text{ J}$
ci2	$W_{AD} \text{ (by electric force)} = 1.0 \times 10^{-8} \text{ J}$
cii	Since $W_{\text{by external force}} = -W_{\text{by electric force}}$, $W_{AB} \text{ (by external force)} = -W_{AB} \text{ (by electric force)} = -1.0 \times 10^{-8} \text{ J}$ Since $\Delta U_{AB} = W_{AB} \text{ (by external force)}$ $q\Delta V_{AB} = -1.0 \times 10^{-8} \text{ J}$ $V_B - V_A = -1.0 \times 10^{-8} / (2.5 \times 10^{-11}) = -400$ $V_B = -400 + 200 = -200 \text{ V}$

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5ci	Taking moments about PO
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